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BACHELOR THESIS



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Modelling exchange rate volatility in Latin America using EGARCH models with points of change.

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Declaration

I hereby declare that I am the sole author of the thesis entitled *Modelling exchange rate volatility in Latin America using EGARCH with points of change*. I duly marked out all quotations. The used literature and sources are stated in the attached list of references.

May 9th, 2022 in Prague

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Signature

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Abstract

This work aims to investigate whether major political events like corruption scandals, presidential impeachment processes or general elections could be associated to points of structural breaks in the volatility of exchange rates in Latin America. Those break points are endogenously determined using the iterated cumulated sums of squares (ICSS) algorithm proposed by Inclan and Tiao (1994).

Since exchange rates are asset prices, finding the effect of shocks in the volatility persistence is important for building accurate asset pricing models. Then, because it better captures asymmetric characteristics of volatility, the EGARCH(1,1) model introduced by Nelson (1991) is used to evaluate whether persistence in volatility is reduced after accounting for those breakpoints.

Keywords: EGARCH models, ICSS algorithm, exchange rate returns, volatility modelling

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1 Introduction

The purpose of this paper is to investigate whether economic and political events can be associated to structural breaks in the exchange rate volatility of Latin American countries and provide proof of the reduction in volatility persistence after these break points are included in the model. The region has been unusually prone to bouts of hyperinflation, exchange rate devaluations, failed currency reforms, banking sector collapses and debt defaults. Moreover, Latin America is frequently subject to shocks, and the frequency and intensity of these shocks - whether positive or negative - could be one reason for the large volatility of economics indicators. It is assumed that exchange rates are good at capturing the effects of shocks, then daily exchange rate returns for five countries in Latin America are included in the study.

The work of Malik (2003) was taken as an inspiration where he examines how major events may lead to sudden changes in volatility as well as the role that shocks play in determining the persistence of volatility over time. A procedure based on an iterated cumulative sums of squares (ICSS) is used to detect the number of sudden changes in variance in the time series, as well as to estimate the time point and magnitude of each detected sudden change in the variance. During periods of increased volatility, the returns have naturally higher standard deviations. Economic and political events surrounding these time points are associated. These volatility shifts are then introduced in an EGARCH model to compute the effect of a given shock on persistence of volatility as well as to test the significance of these points of change. Based on that approach, it is found that the persistence in volatility is reduced.

The rest of the paper is structured as follows. Section 2 briefly explains the topic of persistence in volatility and provides context for the political and economic situation in Latin America that makes it of interest. Section 3 presents the empirical methodology used to determine the points of change in the variance, the persistence in volatility of the series and the methods to evaluate the performance between econometric models. Section 4 introduces the R packages RUGARCH and ICSS. Section 5 describes the data and takes a first look at the differences in volatility. In

Section 6, the empirical results are presented for the five countries, points of change found by the ICSS algorithm are linked to economic and political events, reduction of the persistence in volatility is demonstrated and the performance of the EGARCH model is compared against the GARCH model. Conclusions are drawn in the last section.

2 Background

2.1 Volatility persistence

Financial markets are subject to specific events - oil shocks, wars, financial slumps, changes of policy regimes, natural disasters, pandemics - that have a dramatic impact on modelling financial time series and with the availability of high frequency data for financial markets analysis there has been an increasing number of studies dealing with the persistence of these shocks on the variance of financial instrument returns. Although volatility clearly fluctuates over time, an important question is “how persistent are these changes in volatility following some shock?”

Defining persistence in volatility as the degree to which past volatility explains current volatility, many authors including Aragó and Fernandez-Izquierdo (2003) underline that the degree of persistence has evident economic implications arising from the effect that current volatility has on the predictability of its future value.

There are by now several alternative GARCH type models that attempt to take volatility persistence appropriately into account, implying quite different measures for the conditional volatility. In this paper we consider only two volatility models: the GARCH model and the EGARCH model. The way how persistence is quantified in these models will be specified in the following chapters.

These mentioned shocks which can cause abrupt breaks in the unconditional variance of returns are equivalent to structural breaks in the parameters of the GARCH processes governing the conditional volatility of returns. It is well known that these shocks can bias the estimated persistence of volatility.

Financial market participants and policy makers could be benefited from a better understanding of how shocks can affect volatility over time, especially whether the shocks are persistent or short lived.

2.2 Volatility in Latin America

It is well known that Latin American countries are highly dependent on commodity prices which has helped their economies in reducing poverty and inequality. However,

the political turmoils that followed the collapse of the commodity prices - including oil and copper - in 2014 and the global recession caused by the pandemic in 2020 support the historical evidence that bad economic times often mean bad times for democracy. Increasing levels of unemployment, labor informality, income inequality and poverty very often lead to increasing levels of dissatisfaction with the government and this is because perceptions about the economy are very likely to influence citizens political predisposition.

It is the past economic and political events that took place in Peru, Brazil and Argentina that make them of interest to the author of this work. Chile and Uruguay tend to be considered as relatively developed and stable countries in the region, then it was decided to include them in the study to observe how the volatility of their exchange rate returns behaves next to the ones of Peru, Brazil and Argentina.

2.2.1 Argentina

Argentina is one of those rare countries that were once rich but are now becoming poor. Populist governments in Argentina headed by Cristina Kirchner who became President in December 2007 financed deficit spending social programs and subsidies by extensive foreign borrowing and money printing then driving both inflation and interest rates up.

The center-right Mauricio Macri took offices as the new President in December 2015 on campaign promises to revive an economy crippled by populist social programs and regulations. Unfortunately, the problems left by his predecessor proved to have no easy solution and any effort to roll back public spending and subsidies raised public discontent and met strong political opposition. This situation grew investors' skepticism about the efficacy of Macri administration.

Additionally, the country was hit by a historical drought in the 2017-2018 growing season which dealt a serious blow to Argentina's exports and tax collection, causing its economy to contract.

The government sought financing help from the International Monetary Fund. However, these loans failed to lift the country out of economic crisis and the country slumped

into recession with high inflation. As the Peso plummeted, Argentina's foreign debt became more and more expensive to repay.

2.2.2 Brazil

Since 2014, Brazil has been in the grip of its biggest and most shocking scandal to date, *Petrobras*, which involved the indictment of dozens of high-level businesspeople and politicians as part of a widespread investigation alleging that many millions of dollars had been kicked back to officials of a huge Brazilian oil company - Petrobras - and to high level politicians including the President by prominent Brazilian corporations in return for contracts with Petrobras. Its scope quickly expanded and former presidents and major companies throughout Latin America were implicated in sprawling corruption schemes uncovered by the investigators. Presidents and former presidents of Peru, El Salvador and Panama were also jailed as a result of the investigations.

Millions of people in Brazil took the streets asking for the president to step down, but it only raised the uncertainty because all the government officials that were in line for replacement were also being investigated

With a market capitalization above 30 billion US dollars, Petrobras was the biggest company in South America and one of the biggest producers of oil in the world, then its size amplified the scandal. As it deepened, key infrastructure projects were suspended or scrapped, some suppliers sought bankruptcy protection and job losses mounted by the tens of thousands.

2.2.3 Peru

In 2017, parallel with the impeachment of Brazil's former President Rousseff, legislators in Peru began impeachment proceedings against the President Kuczynski, who was accused of profiting from contracts with the Brazilian construction firm Odebrecht. Kuczynski avoided removal, but few months later in 2018 he offered his resignation after the release of videos that showed key allies trying to buy the support of opposition lawmakers. He was then replaced by his Vice-President Martin Vizcarra.

The following year in 2019, President Vizcarra dissolved Congress arguing that lawmakers were obstructing his anti-corruption agenda. A new Congress was elected in January 2020, but tension remained high between the legislative and the executive. Later in 2020, the newly elected Congress voted to impeach President Vizcarra over allegations he handed out government contracts in return for bribes. Speaker of Congress Manuel Merino assumed office as Peru's third president in four years.

Few days later, Merino resigned after an intense week of protests marked by accusations of police brutality and the killing of two protesters.

Congress voted to replace him with Francisco Sagasti, third president in a week.

3 Methodology

It is known that volatility persistence is typically overestimated if regime shifts are ignored in the conventional GARCH model. However, most studies impose regime shifts based on a priori grounds and thus are subject to the problem that the break dates are correlated with the data.

Imposing regimes in the previously mentioned manner may introduce serious biases into the analysis. To overcome this problem, we detect time periods of sudden changes in variance (i.e., regime shifts) using the ICSS algorithm. Thus, regime shifts are determined endogenously. Moreover, the ICSS method is capable of identifying significant increases and decreases in volatility, so that the beginning and the end of distinct regimes may be identified.

3.1 Detecting points of sudden changes in variance

There are many works aimed at identifying the point of change in a sequence of independent random variables. The approach presented here uses cumulative sums of squares to search for change points systematically at different pieces of the series. It is based on a centered version of the cumulative sum of squares presented by Brown, Durbin, and Evans (1975).

Inclan and Tiao (1994) consider series that present a stationary behavior for some time, then suddenly the variance of the error term changes; it stays constant again for some time at this new value, until another change occurs.

The methodology used in detecting sudden discrete changes in variance is based on the ICSS algorithm given by Inclan and Tiao (1994). Their analysis focuses on detecting a variance change in a given time series due to a sudden shock that changes the variance until the series experiences a future shock. Let ε_t , $t = 1, \dots, T$ be a series with zero mean and with unconditional variance σ_t^2 . Let the variance within each interval be given by τ_t^2 , $j = 0, 1, \dots$, N_T is the total number of variance changes in T

observations, and $1 < \kappa_1 < \kappa_2 < \dots < \kappa_{NT}$ are the change points,

$$\sigma_t^2 = \tau_0^2 \text{ for } 1 < t < \kappa_1 \quad (1a)$$

$$\sigma_t^2 = \tau_1^2 \text{ for } \kappa_1 < t < \kappa_2 \quad (1b)$$

$$\sigma_t^2 = \tau_{NT}^2 \text{ for } \kappa_{NT} < t < T \quad (1c)$$

In order to determine the number of changes in variance and the time point of each variance shift, a cumulative sum of squares is used. Let us denote $C_k = \sum_{t=1}^k \varepsilon_t^2$, $k = 1, \dots, T$, as the (mean-centered) cumulative sum of squares from the first observation to the k th point in time. Define the D_k statistic as follows:

$$D_k = (C_k/C_T) - k/T, \quad k = 1, \dots, T \text{ with } D_0 = D_T = 0. \quad (2)$$

If the series contains no changes in variance then the D_k statistic will oscillate around zero and if plotted against k will look like a horizontal line. However, if the series contains changes in variance, then it will plot as a drift from zero. Significant changes in variance are detected using critical values obtained from the distribution of D_k under the null hypothesis of constant variance. The null hypothesis of homogeneous variance can be rejected if the maximum absolute value of D_k is greater than the critical value. Let k^* denote the value at which $\max_k |D_k|$ is reached. If $\max_k \sqrt{(T/2)}|D_k|$ falls outside the predetermined boundary, then k^* is taken as the time point of a variance change in the series the term $\sqrt{(T/2)}$ is required to standardize the distribution.

Aggarwal, Inclan, and Leal (1999) use critical value of 1.36, which is the 95th percentile of asymptotic distribution of $\max_k \sqrt{(T/2)}|D_k|$. Therefore the upper and lower boundaries are set at ± 1.36 . A change point in variance is identified if it falls outside these boundaries.

The original ICSS algorithm used in the previous studies is designed for processes that are i.i.d., which is a very strong assumption for financial data, in which there is evidence of conditional heteroskedasticity. Sansó et al. (2004) show that the size distortions are important for heteroskedastic conditional variance processes from Monte Carlo simulations. Their results thus invalidate in practice the use of the

original ICSS algorithm for financial time series. To overcome this problem, the authors propose a new test that explicitly considers the fourth moment properties of the disturbances and the conditional heteroskedasticity.

As mentioned earlier, the ICSS algorithm assumes constant variance within regime. Because financial data are documented to have time varying variance, we amend the ICSS methodology as suggested by Sanso, Arago, and Carrion (2004). They propose that the critical value of 1.4058 (i.e., correcting for kurtosis), generated via Monte Carlo simulations, should be used when using the ICSS algorithm on financial data. In this work, their suggestion is followed and the critical value of 1.4058 is used as it explicitly accounts for conditional heteroscedasticity. Failure to properly adjust the critical value will result in over-rejection of the null hypothesis when the standard ICSS algorithm is used to detect breakpoints on conditionally heteroscedastic data. In other words, the researcher is likely to find more spurious structural breaks if critical values are not properly adjusted.

Sanso, Arago, and Carrion (2004) validate their claim using Monte Carlo simulations and provide examples based on real world data.

However, Inclan and Tiao (1994) document via simulations that it may be difficult for the D_k statistic to detect multiple variance changes when examining the entire series simultaneously. For example, a major break point followed by a moderate-sized break point may be overlooked by the D_k statistic. To overcome this 'masking effect', they propose an algorithm that looks at different pieces of the series for identification of change points in variance. The ICSS algorithm works by looking for one break point at a time through the D_k statistic. If a break point is detected, then the sample is further segmented to find other break points.

3.1.1 R Package *ICSS*

The package implements the Iterative Cumulative Sum of Squares (ICSS) algorithm by Inclan and Tiao (1994) which detects structural breakpoints in the variance of time series data. This package allows the user to call two functions : *data* and *ICSS*.

1. *data* function is a generator of random data that follows the scheme proposed in Inclan and Tiao (1994).
2. *ICSS* function returns a numerical vector containing the location of structural breakpoints or NA if none breakpoints are found.

For the purposes of this work, only the *ICSS* function will be used to get the vector of breakpoints. Certainly additional steps need to be performed in order to have the list of breakpoints in a format suitable to be used in the next steps of our work.

3.2 GARCH Models

Various problems such as option pricing in finance have motivated the study of the volatility of a time series. In ARMA framework, current values of the series of interest is written as a linear function of its own lagged values and current past noise process or innovations. These innovations are assumed to be independent and identically distributed, or, technically speaking, homoskedastic in nature. Even though these models are useful to model the conditional mean of a process when the conditional variance is constant, they are not suitable for nonlinear financial and monetary problems where the assumption of a constant conditional variance is usually violated.

Introduced by Engle (1982), the ARCH/GARCH specifically model and forecast conditional variances. These models and their variants can successfully capture the stylized facts of the volatility of stock returns, specially volatility clustering, long memory, leptokurtosis and the leverage effect. The ARCH model assumes that the variance of the current period is an equally weighted average of the squared residuals of the previous days.

Bollerslev (1986) extends the ARCH model to GARCH, or the generalized ARCH which by using declining weights for the squared residuals that are estimated by the model provides a more flexible framework to capture various dynamic structures of conditional variance.

In these problems, we are concerned with modeling the return or growth of a series. For example, if x_t is the value of an asset at time t , then the relative gain or return, r_t , of the asset at time t is

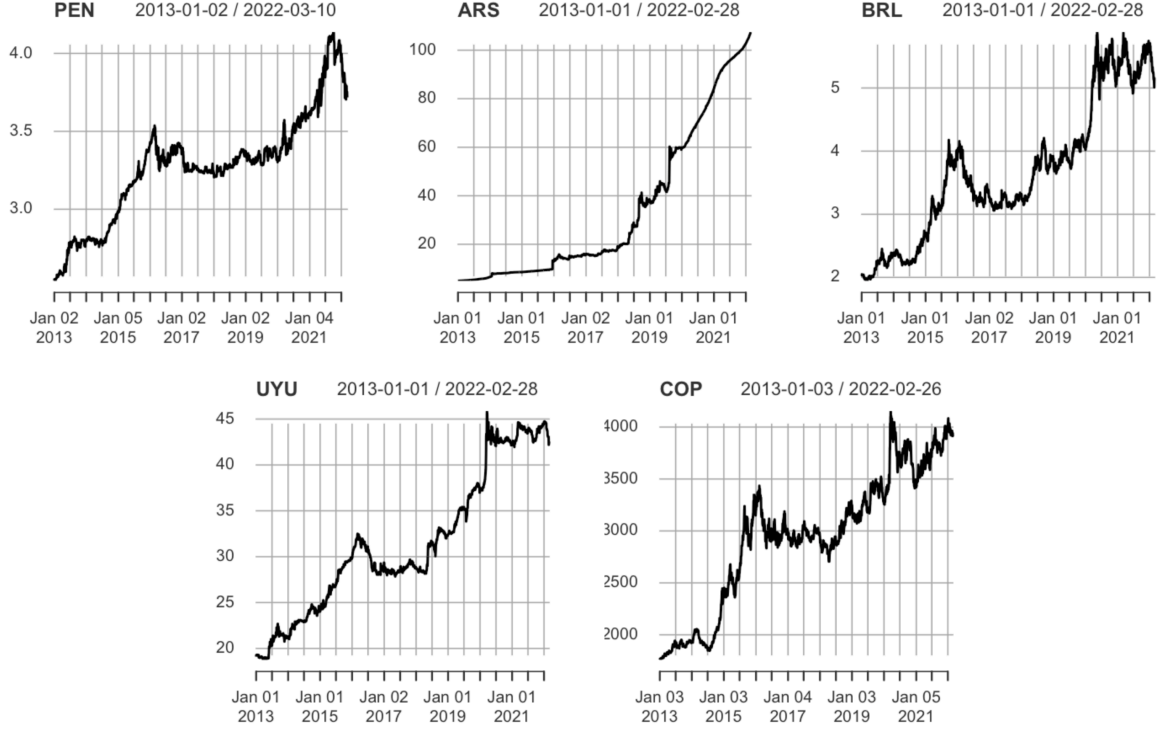


Figure 1: Time series of exchange rates

$$r_t = \frac{x_t - x_{t-1}}{x_{t-1}}. \quad (3)$$

Figure 1 shows the time series of the exchange rates while Figures 2 to 6 show the time series of the returns.

3.2.1 GARCH(1,1)

The conditional variance equation of the standard GARCH(1,1) model has the following form:

$$\sigma_t^2 = \omega + \beta\sigma_{t-1}^2 + \alpha\varepsilon_{t-1}^2, \quad (4)$$

where ω , α , β must be non-negative parameters with $\alpha + \beta < 1$ and ε_t denote the error term.

3.2.2 Stationarity

An important consideration of GARCH models is whether the conditional variance σ_t^2 converges to a stationary distribution. The GARCH(1,1) equations have a stationary solution if and only if $\alpha + \beta < 1$. This is:

$$\sigma_t^2 = \frac{\omega}{1 - \alpha - \beta}. \quad (5)$$

The sum of α and β quantifies the persistence of shocks to conditional variance.

3.2.3 EGARCH(1,1)

Standard GARCH models assume that positive and negative error terms have a symmetric effect on the volatility. In other words, good and bad news have the same effect on the volatility in this model. In practice this assumption is frequently violated in that the volatility increases more after bad news than after good news. This is commonly referred as the leverage effect according to which negative shocks tend to have more impact on volatility than positive shocks of the same magnitude. Therefore, the result based on the GARCH(1,1) models may be questionable because it does not take into account the asymmetry and non-linearity of the conditional variance. To remedy these shortcomings, it seems more appropriate to apply an asymmetric GARCH model.

In the exponential GARCH or EGARCH model proposed by Nelson (1991), the log-volatility is expressed as a linear combination of its past values and past values of the positive and negative parts of the innovations. Two main reasons for the success of this formulations are that (i) it allows for asymmetries in volatility and (ii) it does not impose any positivity restrictions on the volatility coefficients.

Thus, we estimate Nelson's EGARCH(1,1) model or exponential GARCH(1,1) according to the following formulation:

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}}, \quad (6)$$

where α determines the sign effect and γ determines the size effect.

Another advantage of using EGARCH is that the positivity of the parameters is guaranteed since we are working with the log of the variance. Also, there are no restrictions on the parameters ω , α and γ . However, to maintain stationarity, β must be positive and less than 1. The leverage effect is indicated by the value of γ .

In this work, EGARCH(1,1) models are used to model the volatility of the exchange rates.

3.3 Combined model of sudden changes with GARCH

The GARCH model is one way of capturing the persistence of volatility observed in a time series. The model can be modified to incorporate sudden changes in the variance also.

Lastrapes (1989) and Lamoureux and Lastrapes (1990) argue that persistence in volatility tends to be overestimated when standard GARCH models are applied to a series with underlying sudden changes in variance. These shifts in volatility could be due to regime/structural changes that are caused by political, social or economic events.

We consider a model that combines the shifts in variance with GARCH. Our empirical approach first detects the change points by using the ICSS algorithm and then dummy variables are introduced into the variance equation of the GARCH model to account for the sudden changes in variance. The present study uses an EGARCH(1,1) model with sudden changes in variance, which were identified by using ICSS algorithm.

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + d_1 D_1 + \dots + d_n D_n, \quad (7)$$

where $D_1, \dots, D_n$ are the dummy variables taking a value of one from each point of sudden change of variance onwards, zero elsewhere.

The impact on the GARCH parameters of including the dummy variables in the equation is discussed in the following sections.

3.4 Evaluating the models

Introduced by Hirotugu Akaike in 1973 as an extension to the maximum likelihood principle, the Akaike Information Criterion (AIC) combines into a single procedure the process of estimation with structural and dimensional determination. It is a single number score that will be used to determine which of multiple models is most likely to be the best model for a given dataset. It works by evaluation the model's fit on the training data, and adding a penalty term for the complexity of the model. The desired result is to find the lowest possible AIC.

AIC is computed as follows:

$$AIC = 2K - 2\ln(L) \tag{8}$$

where K is the number of model parameters and $\ln(L)$ is the log-likelihood of the model.

From several models of a given data set, the best model is the one which has the lowest AIC value.

In this work, the AIC will be used to compare the EGARCH and GARCH models as well as to compare the models including and excluding the dummy variables that contain the information about break points.

4 Data

The data used in this study are taken from January 2013 to March 2022 and includes about 2389 observations per country. Nominal exchange rates (against US dollar) for Argentine peso, Uruguayan peso, Brazilian real, Colombian peso and Peruvian sol are used in this work. All data were obtained from the Wall Street Journal website. Table 1 provides descriptive statistics for all daily return series used in the study.

The pikes in the series are no mistake in the data set. They respond to major events that took place in each country for example the one on December 2017 when Argentina lifted capital controls.

Note how despite sharing similar characteristics and been all subjected to global events like fluctuations in the commodity prices, each country's currency follows a different story.

Mid-2013 was chosen as starting point given it was a relatively stable year for the countries included in the study and is before some major global events like the COVID-19 pandemic that should be detected as points of change in the variance.

Table 1: Descriptive Statistics

Currency	Mean	Standard deviation	Skewness	Kurtosis
Peruvian sol	0.000177	0.003901	-0.3318	12.1469
Argentine peso	0.001353	0.011670	14.8892	395.6401
Brazilian real	0.000438	0.010179	0.1144	5.7320
Uruguayan peso	0.000348	0.004980	1.2740	15.7758
Colombian peso	0.000398	0.007729	0.3581	7.0880

5 Empirical results

5.1 Descriptive Statistics

One clear indication from Table 1 is that all series are leptokurtic (i.e. fat tails), a feature that has been widely documented in the literature for many financial time series. A large kurtosis is associated with a high risk for an investment because it indicates high probability of extremely large or extremely small returns.

Peruvian sol has the smallest standard deviation among all five cases while Argentine peso has the largest standard deviation. Peruvian sol was the only negatively skewed series. Argentina's highly positive skew means it is safer to hold money in US dollars than in domestic currency.

5.2 ICSS breakpoints

The ICSS algorithm results are presented in Table 2. As can be seen in the table, the number of sudden changes is higher for series with a lot of spikes like Uruguay. It will be seen later that not all those 23 break points are significant.

Note there are some common break points between the countries. One example is the COVID-19 outbreak. The only country where the COVID-19 outbreak was not recognized as a point of change is Argentina. Argentina presents a spike in the beginning of April 2020, but this spike is very small compared to the turbulent times it went through between 2016 and 2020.

While some points of change are linked to external shocks, political events and

economic decisions also represent a big portion of the total number of breakpoints. In some cases, break points due to external shocks are followed by break points due to political events then supporting the work of authors who state that economic downturns lead voters toward anti-incumbent preferences.

While sometimes it is the direct effect of these shocks that created volatility, in many cases it is simply the fear of them that leads the authorities to create precautionary recessions or the private sector to speculate on their potential arrival. This is the case of Peru in April 2020 when a radical left-wing candidate wins the first round in the presidential runoff and shakes the financial markets even when his final victory had not yet even happened.

About the COVID-19 break point, see how even though the first infection was announced around the same time for Brazil, Peru, Uruguay and Colombia, the next break point is different from country to country and the reason can be found in how each country managed the pandemic plus the domestic political and social conflicts that arose as a result of the pandemic.

In Peru, the measures taken by the central government generated such a discontent in the population leading to a political crisis that ended with the dissolution of the Congress, impeachment of the President and protests all over the country. The end of this political crisis coincides with the break point identified by the ICSS algorithm when on November 24th 2020 the Peruvian Congress chose a centrist statesman and consensus candidate to become the country's third president in a week sending a message of stability to the markets which can be associated to the lower volatility till the next break point.

The case of Uruguay is also interesting. Even though it is considered one of the most developed and stable countries in the region, the Uruguayan peso exchange rate is full of spikes which cannot be associated to major political events yet those spikes cause the ICSS algorithm to overestimate the amount of points of change in the variance.

The break points in Brazil are in majority associated to the (i) *Petrobras* scandal and (ii) commodity price fluctuations mentioned in Section 1: Impeachment processes,

allegations against high-level politicians, fraud investigations, speculation in the media, projects and political activity paralysed, anti-government protests and voters struggling to find anyone to believe in.

Be aware that markets may well anticipate some events in advance and respond accordingly, as well as possibly taking some time for other events to actually take effect, it is not expected these events to correlate to changes in sudden variance on any particular or specific day. Rather, they simply highlight the idea that major events, crises, changes in policy, and so forth may be a contributing factor in the sudden change.

Table 2: Sudden changes in volatility.

Currency	No. of change points	Time period	Standard deviation	Events
Peruvian sol	21	February 27, 2020 - April 28, 2020	0.0082	COVID-19 outbreak
		November 24, 2020 - April 05, 2021	0.0023	Congress elects new President after political crisis
		April 05, 2021 - August 02, 2021	0.0097	Radical left-wing candidate wins the first round in the presidential runoff
Argentine peso	6	December 17, 2015 - May 02, 2018	0.0163	Currency controls lifted
		May 02, 2018 - August 12, 2019	0.0189	President Macri announces new IMF debt deal
		December 27, 2019 - April 06, 2021	0.0019	Alberto Fernandez swears as new President of Argentina
Brazilian real	11	October 07, 2014 - April 20, 2016	0.0127	Collapse in global commodity prices
		April 02, 2019 - March 09, 2020	0.0073	Brazil swears in far-right President Jair Bolsonaro
		March 09, 2020 - September 29, 2020	0.0155	Sitting Brazilian President faces charges amid corruption scandals
		March 09, 2020 - September 29, 2020	0.0155	COVID-19 outbreak
Uruguayan peso	23	February 28, 2020 - April 10, 2020	0.0137	COVID-19 outbreak
Colombian peso	7	November 29, 2014 - December 02, 2016	0.0109	Oil price plunge
		February 28, 2020 - March 20, 2020	0.0239	COVID-19 outbreak

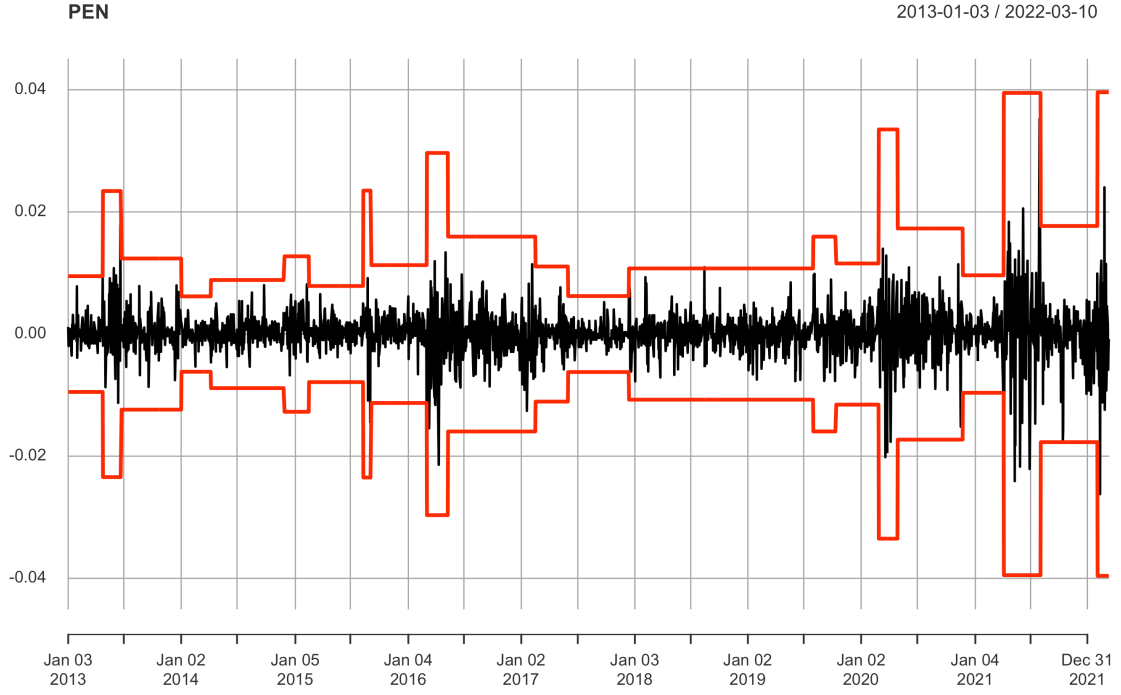


Figure 2: Peruvian Sol

5.3 Combined model of sudden changes with EGARCH

The next step was to incorporate these break points to the standard EGARCH model. A set of dichotomous variables corresponding to the sudden points of change is constructed to account for the different volatility regimes.

The results are presented in the following way:

1. Table 3 presents the scores of the Akaike Information Criterion (AIC) after running the regressions with and without the dummy variables in the GARCH(1,1) and EGARCH(1,1) models.
2. Table 4 contains the estimates and p-values of the parameters alpha, beta and gamma for the EGARCH(1,1) models with and without the dummy variables.
3. Tables 5 to 9 contains the estimates and p-value of the points of change in the variance. One country per table. These tables can be found in the Appendix.

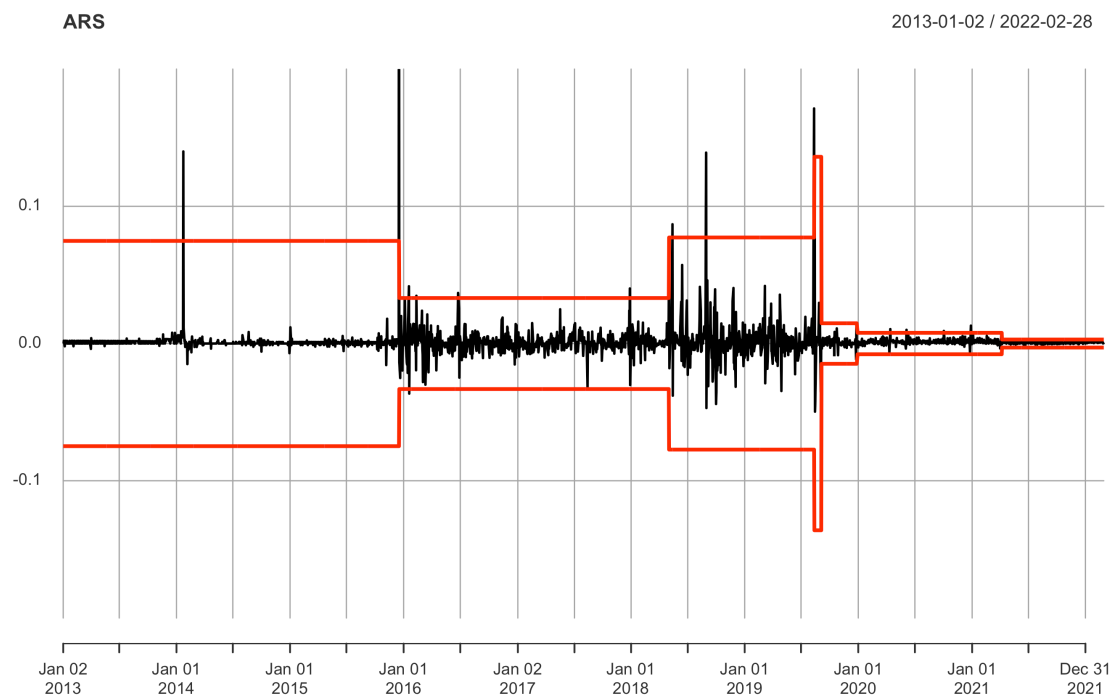


Figure 3: Argentine peso

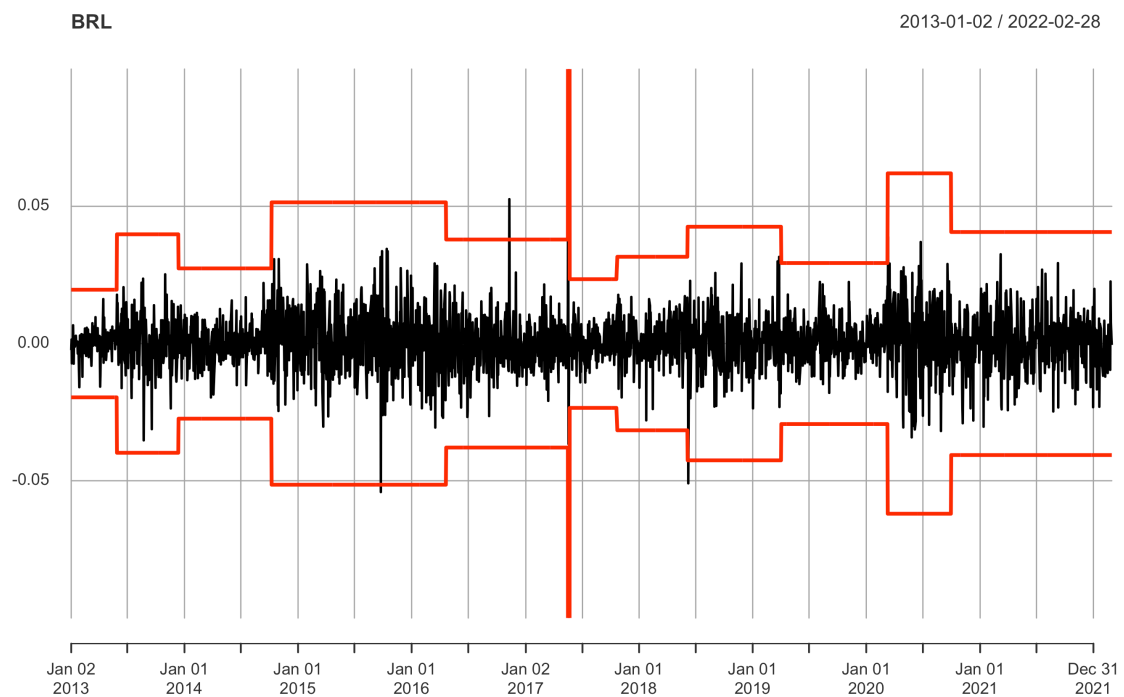


Figure 4: Brasil real

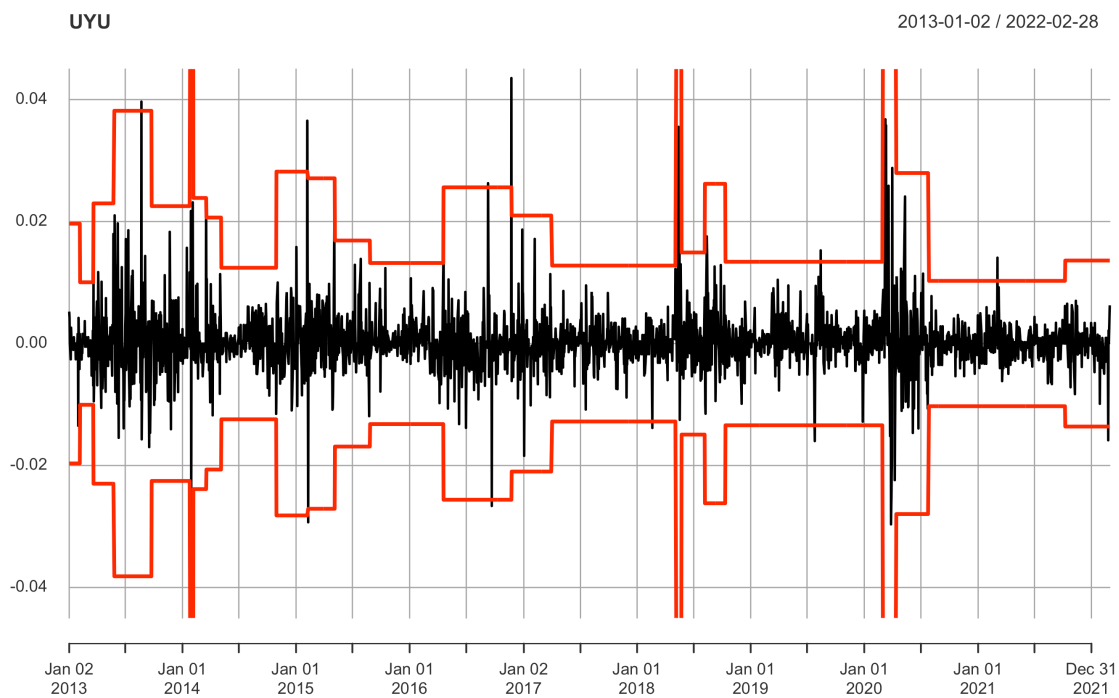


Figure 5: Uruguayan peso

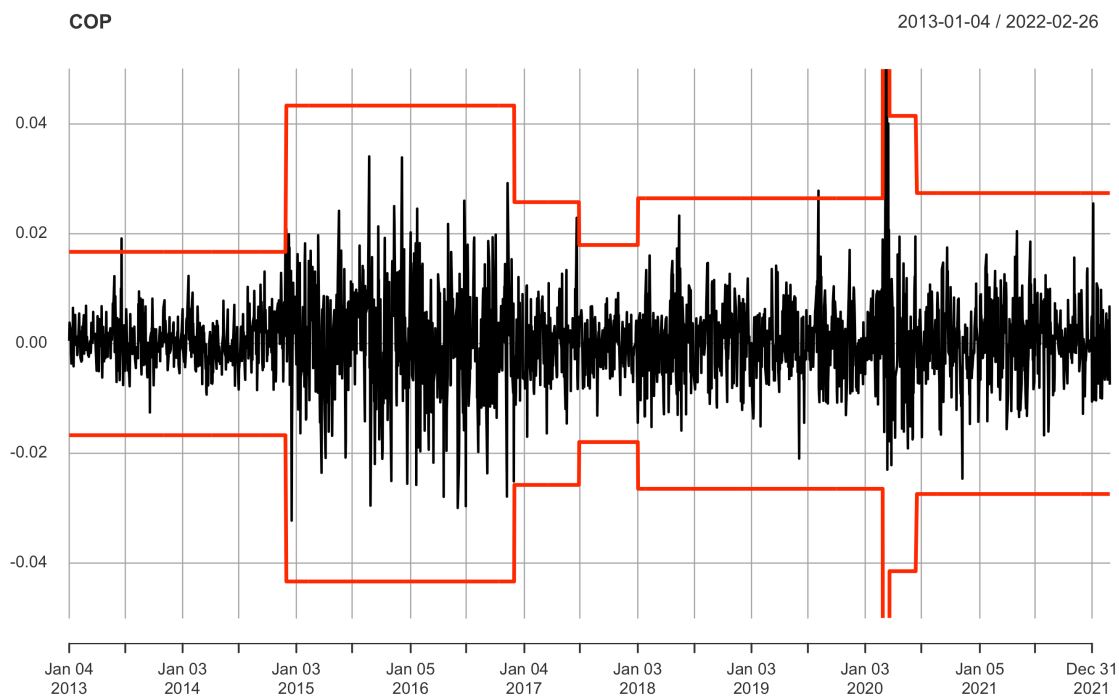


Figure 6: Colombian peso

The tables follow the previously described layout so the reader can focus (i) first in comparing the performance of the EGARCH models next to the GARCH models, (ii) then on the reduction of the persistence in volatility after accounting for points of change in the variance and last (iii) in the effect and significance of each particular break point.

5.3.1 GARCH(1,1) vs EGARCH(1,1)

Note in Table 3 that for all the economies included in the study, the values of the AIC tell us that the EGARCH(1,1) models with dummy variables are better at modelling the volatility of the exchange rate returns.

Also, in all countries except for Peru and Uruguay, the simple GARCH model with points of change has a lower AIC than the GARCH model without the points of change. However, if we remember that Peru and Uruguay are the countries with the largest number of points of change - 21 and 23 - and that the Akaike Information Criterion is designed to penalize the complexity of the model, then it makes sense that models with more than twenty additional explanatory variables will be highly penalized next to the simple GARCH(1,1) without dummy variables.

Furthermore, from Tables 5 through 9 it can be highlighted that while none of the breakpoints are significant when included in the GARCH models, their inclusion is significant when modelling volatility with the EGARCH models.

Because there seems to be enough evidence in this work that EGARCH models are better at modelling the volatility of exchange rate returns than GARCH models, Table 4 focuses in the parameters and p-values of the EGARCH models.

Table 3: EGARCH(1,1) parameters with and without dummy variables for sudden changes in variance

Currency	Without dummy variables		With dummy variables	
	GARCH	EGARCH	GARCH	EGARCH
Peruvian sol	-8.7722	-8.7717	-8.7556	-8.8304
Argentine peso	-8.7566	-8.6341	-8.7607	-8.8042
Brazilian real	-6.4774	-6.4963	-6.4720	-6.5251
Uruguayan peso	-8.3676	-8.3690	-8.3472	-8.4090
Colombian peso	-7.1742	-7.1822	-7.1742	-7.2105

5.3.2 Accounting for sudden changes in variance.

In our analysis, the sudden changes in variance detected by the ICSS algorithm were incorporated directly into an EGARCH model i) to test the relative strengths of each break point and ii) to analyze the impact of including the break points on the persistence of variance.

Remembering that γ is the asymmetry effect and looking at Table 4 it can be found that in the simple EGARCH(1,1) model - this is before including break points - there is not enough evidence to reject the asymmetric effect in all economies except for Argentina, but this situation changes after dummy variables are included in the model. However, it also changes for Brazil and Colombia where the asymmetric effect is not significant anymore.

In the case of Brazil, the three parameters - α , β and γ - become non-significant after accounting for points of change in the variance, but at the same time, the effects of each of the eleven points of change in Brazil reveal to be significant and since the AIC indicated that EGARCH(1,1) with dummy variables was better than the other models, then it could even be argued that the points of change alone are better at modelling volatility for Brazil.

As demonstrated by Lamoureux and Lastrapes's (1990), when regime shifts are incorporated directly into a GARCH model, the persistence of variance indicated by the GARCH model should decrease dramatically. This work aims to demonstrate the same but using an EGARCH model instead.

Now, persistence in volatility is measured as the sum of α and β when dealing with GARCH models, but for EGARCH models it is measured as the value of β alone. The results presented in Table 4 clearly show the decrease in persistence in volatility when break points are accounted for in the model. The economy with the largest decrease is Brazil where not only the value of the parameter β jumped from 0.9946 to 0.0714, but the persistence in volatility is not significant anymore in the EGARCH model with the dummy variables. The same happens for Peru and Argentina. There is not enough evidence to reject the significance of the persistence in volatility in

Uruguay and Colombia - reflected by the parameter β -, but its decrease has been demonstrated.

In most of the series included in this work, the β is close to one before including the break points, implying extreme persistence in volatility. Brazil followed by Colombia have the highest persistence in the EGARCH(1,1) model without dummy variables. This ranking changed after the break points are included in the model as Colombia goes to the top with the highest persistence. A possible explanation could be found in the significance of the break points. As can be seen in Table 8 and Table 9, none of the break points found in Colombia are significant while in Brazil it is the opposite.

Table 4: EGARCH(1,1) parameters with and without dummy variables for sudden changes in variance

Currency	EGARCH(1,1)			EGARCH(1,1) with dummy variables		
	α	β	γ	α	β	γ
Peruvian sol	0.0106 (0.6672)	0.9602 (0.0000)	0.3835 (0.0000)	-0.0264 (0.3990)	0.2039 (0.2723)	0.2927 (0.0000)
Argentine peso	-0.3029 (0.4109)	0.8429 (0.0000)	1.7301 (0.1455)	-0.0287 (0.6790)	0.2684 (0.1717)	0.5596 (0.0000)
Brazilian real	0.0495 (0.0000)	0.9946 (0.0000)	0.0645 (0.0000)	0.0312 (0.4073)	0.0714 (0.6371)	0.0262 (0.6038)
Uruguayan peso	0.0045 (0.8572)	0.9642 (0.0000)	0.3070 (0.0000)	-0.0238 (0.4948)	0.4872 (0.0000)	0.3609 (0.0000)
Colombian peso	0.0584 (0.0001)	0.9845 (0.0000)	0.1700 (0.0000)	0.0644 (0.0726)	0.8107 (0.0000)	0.1382 (0.0565)

6 Conclusions

This study investigates sudden shifts in volatility and re-examines volatility persistence for the exchange rate returns of five countries in Latin America and the events that are associated with the increased volatility. Sudden shifts are largely explained by domestic and international economic and political factors. These include the currency controls in Argentina, corruption scandals in Brazil and the rise of the left in Peru.

Furthermore, it was proved that exponential GARCH models generally outperform simple GARCH models because they capture the asymmetric characteristics of financial time series. Also, when in the group of simple GARCH models, the model without dummy variables is compared to the one including them, it can happen that the models with a large amount of detected points of change have a lower AIC score. And this is because the AIC penalizes the complexity of the model.

There are no conclusive results about whether it is domestic political events or external economic shocks that have the strongest effect, but as mentioned in the beginning of this work: bad economic times often mean bad times for democracy. Then it is not that easy to isolate political events from economic events and this insight will be left for future studies.

It was shown that incorporating points of change in the EGARCH(1,1) model significantly reduces volatility persistence. In some cases volatility persistence is not significant anymore after the incorporation of the dummy variables given all the effect is now captured by the break points. These results are consistent with earlier research that stated that ignoring regime shifts in the presence of sudden changes in variance can lead to drastically different and erroneous conclusions about the degree of volatility persistence that exists in shock returns.

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Appendix

Table 5: Breakpoints for Peruvian sol when points of change are included in the model

Breakpoint	GARCH	EGARCH	Breakpoint	GARCH	EGARCH
2013-04-24	0.0000 (0.8826)	1.9916 (0.0001)	2017-06-01	0.0000 (0.9950)	-0.8755 (0.0527)
2013-06-21	0.0000 (0.9117)	-1.3726 (0.0008)	2017-12-14	0.0000 (0.9087)	0.8260 (0.0036)
2014-01-02	0.0000 (0.9271)	-1.0941 (0.0038)	2019-07-31	0.0000 (0.9548)	0.6958 (0.0137)
2014-04-07	0.0000 (0.8656)	0.6606 (0.0103)	2019-10-11	0.0000 (0.9679)	-0.4241 (0.1215)
2014-11-28	0.0000 (0.9279)	0.7650 (0.0014)	2020-02-27	0.0000 (0.9772)	1.7359 (0.0000)
2015-02-17	0.0000 (0.9584)	-1.0002 (0.0018)	2020-04-28	0.0000 (0.9835)	-1.0805 (0.0000)
2015-08-11	0.0000 (0.9580)	1.8825 (0.0014)	2020-11-24	0.0000 (0.9907)	-1.2748 (0.0009)
2015-09-04	0.0000 (0.9685)	-1.4502 (0.0064)	2021-04-05	0.0000 (0.9902)	2.5082 (0.0000)
2016-03-03	0.0000 (0.9818)	1.9725 (0.0004)	2021-08-02	0.0000 (0.9937)	-1.4631 (0.0002)
2016-05-10	0.0000 (0.9850)	-1.1092 (0.0027)	2022-02-01	0.0000 (0.9982)	1.6963 (0.0001)
2017-02-16	0.0000 (1.0000)	-0.7391 (0.0447)			

Table 6: Breakpoints for Uruguayan peso when points of change are included in the model

Breakpoint	GARCH	EGARCH	Breakpoint	GARCH	EGARCH
2013-02-05	0.0000 (0.7897)	0.0822 (0.8351)	2016-04-18	0.0000 (0.9793)	0.4808 (0.0382)
2013-03-20	0.0000 (0.8145)	1.2539 (0.0010)	2016-11-23	0.0000 (0.9821)	0.0624 (0.7578)
2013-05-24	0.0000 (0.8673)	0.6207 (0.0341)	2017-03-31	0.0000 (0.9845)	-0.4952 (0.0129)
2013-09-23	0.0000 (0.9045)	-0.6807 (0.0163)	2018-05-04	0.0000 (0.9948)	1.7694 (0.0000)
2014-01-24	0.0000 (0.9243)	1.6734 (0.0000)	2018-05-23	0.0000 (0.9956)	-1.3372 (0.0000)
2014-02-04	0.0000 (0.9241)	-2.1113 (0.0000)	2018-08-06	0.0000 (0.9987)	0.5569 (0.0117)
2014-03-18	0.0000 (0.9166)	0.5140 (0.1404)	2018-10-11	0.0000 (1.0000)	-0.9785 (0.0007)
2014-05-05	0.0000 (0.9172)	-0.6911 (0.0508)	2020-02-28	0.0000 (0.9923)	1.8884 (0.0000)
2014-10-29	0.0000 (0.9374)	0.7844 (0.0082)	2020-04-10	0.0000 (0.9936)	-0.9104 (0.0001)
2015-02-06	0.0000 (0.9607)	0.1430 (0.2909)	2020-07-24	0.0000 (0.9954)	-1.0857 (0.0001)
2015-05-05	0.0000 (0.9723)	-0.5426 (0.0049)	2021-10-07	0.0000 (0.9948)	0.1269 (0.4942)
2015-08-26	0.0000 (0.9823)	-0.4214 (0.0388)			

Table 7: Breakpoints for Argentine peso when points of change are included in the model

Breakpoint	GARCH	EGARCH
2015-12-17	0.0000 (0.9992)	2.0136 (0.0006)
2018-05-02	0.0000 (0.9978)	0.8839 (0.0028)
2019-08-12	0.0000 (0.9996)	1.5967 (0.0428)
2019-09-04	0.0000 (0.9996)	-3.5193 (0.0042)
2019-12-27	0.0000 (0.9998)	-0.8201 (0.0287)
2021-04-06	0.0000 (1.0000)	-1.0075 (0.0005)

Table 8: Breakpoints for Colombian peso when points of change are included in the model

Breakpoint	GARCH	EGARCH
2014-11-29	0.0000 (0.0014)	0.3698 (0.2341)
2016-12-02	0.0000 (0.9999)	-0.1899 (0.2401)
2017-06-28	0.0000 (0.9983)	-0.1282 (0.2989)
2018-01-03	0.0000 (0.9485)	0.1207 (0.3081)
2020-02-28	0.0000 (0.9693)	0.6497 (0.1444)
2020-03-20	0.0000 (0.9896)	-0.4584 (0.1004)
2020-06-13	0.0000 (0.9961)	-0.1657 (0.2885)

Table 9: Breakpoints for Brazilian real when points of change are included in the model

Breakpoint	GARCH	EGARCH
2013-05-28	0.0000 (0.1529)	1.4226 (0.0000)
2013-12-12	0.0000 (0.8317)	-0.7095 (0.0001)
2014-10-07	0.0000 (0.6211)	1.2166 (0.0000)
2016-04-20	0.0000 (0.9780)	-0.6198 (0.0004)
2017-05-17	0.0000 (0.9968)	3.1819 (0.0000)
2017-05-22	0.0000 (1.0000)	-4.0139 (0.0000)
2017-10-20	0.0000 (0.6572)	0.4825 (0.0195)
2018-06-05	0.0000 (0.8626)	0.5239 (0.0090)
2019-04-02	0.0000 (0.9320)	-0.6470 (0.0017)
2020-03-09	0.0000 (0.8630)	1.4989 (0.0000)
2020-09-29	0.0000 (0.9541)	-0.8715 (0.0001)